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AI 100

2-27-26

Problem Definition:

The objective of this project was to solve a classification problem using a deep learning model. I chose to solve for if a deep learning model was capable of predicting whether a golf shot will be a good shot or a bad shot. This was solved using an MLP as my choice of a deep learning model. As a Professional Golf Management student, this problem connects directly to performance analysis commonly used in modern golf instruction using launch monitor technology.

Dataset Curation and Description:

Since publicly available launch monitor data of such a large scale is very limited, I created a synthetic dataset that mimics realistic golf shot performance. The dataset includes variables which are commonly measured by launch monitors. This includes club speed, ball speed, smash factor, spin rate, and a few others. The dataset contains 1,200 simulated golf shots that are labeled as either a good shot (1) or a poor shot (0). The shot quality was determined using performance based metrics like higher smash factor, optimised launch conditions, longer carry distance, and tighter dispersion increased the chance of a good shot outcome. The final dataset was saved as a CSV file and used to evaluate the accuracy of the MLP.

Deep Learning Model:

I used a Multi Layer Perceptron to analyze the data and predict the shot result. This was implemented using TensorFlow and Keras to perform binary classification. All numerical features were standardized using feature scaling to improve the accuracy. The Neural network consisted of the following:

- Input layer containing golf performance variables
- Dense hidden layer with 64 neurons using ReLU activation
- Dropout layer to reduce overfitting
- Second dense layer with 32 neurons
- Output layer with a sigmoid activation function producing the probability of a good shot

The model was trained using the Adam optimizer and binary cross-entropy loss function. Early stopping was applied based on validation loss to prevent overfitting during training.

Results & Performance:

The trained neural network achieved a test accuracy of 61.67%. The confusion matrix shows that the model correctly identified 92 good shots and 19 poor shots. This shows that the model was better at identifying good shots compared to poor shots, which I would assume is influenced by the ratio of good to poor shots in the dataset. Some misclassification had to occur, but the model showed that it understood and learned meaningful relationships between swing variables and shot outcomes. Results shown below:

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TEST ACCURACY: 0.6167

CONFUSION MATRIX:
 [[19 47]
 [22 92]]

CLASSIFICATION REPORT:
              precision    recall  f1-score   support

           0     0.4634    0.2879    0.3551        66
           1     0.6619    0.8070    0.7273       114

    accuracy                         0.6167       180
   macro avg     0.5626    0.5474    0.5412       180
weighted avg     0.5891    0.6167    0.5908       180
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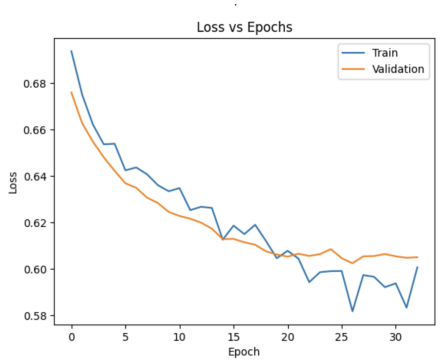
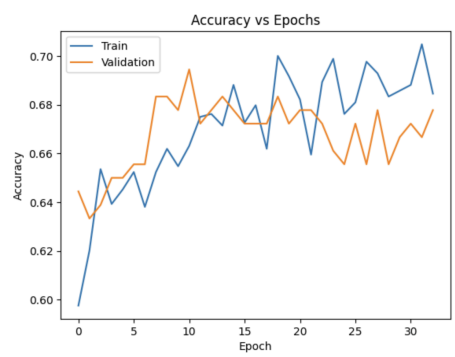
Looking at the graphs, training and validation accuracy increased over time while loss decreased across epochs, indicating that the network successfully learned patterns within the dataset. Early stopping helped present excessive training once validation performance stabilised, which then helped reduce overfitting.





Lessons & Experience:

This project demonstrated how deep learning models can be applied to real world sports performance. One major takeaway was the importance of preparing data. By preprocessing and particularly feature scaling, it significantly improved training stability. I also learned that neural networks can capture non linear relationships

between variables such as launch conditions and shot outcomes. This project also highlighted how dataset balance impacts the performance. This was shown as the model performed better predicting good shots compared to poor shots due to the dataset having more good shots in it. In conclusion, this project successfully demonstrated the application of deep learning model (MLP) to classify golf shot performance. My understanding of deep learning workflows and dataset creation through model evaluation is stronger than ever!